Q1: 24 Feb (Shift 2) - Numerical

Assuming ideal behaviour, the magnitude of log K for the following reaction at $25^{\circ}C$ is $x \times 10^{-1}$. The value of x is ____.(Integer answer)

$3HC \equiv CH_{(g)} \rightleftharpoons C_6H_{6(\ell)}$

$\left[\text{Given : } \Delta_f G^{\circ}(HC \equiv CH) = -2.04 \times 10^5\right] \text{mol}^{-1}; \Delta_f G^{\circ}(C_6H_6) = -1.24 \times 10^5 \text{ J mol}^{-1}$
$\left. R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}\right]$

Q2: 25 Feb (Shift 1) - Numerical

The reaction of cyanamide, $NH_2CN_{(s)}$ with oxygen was run in a bomb calorimeter and ΔU was found to be $-742.24 \text{ kJ mol}^{-1}$. The magnitude of ΔH_{298} for the reaction

$NH_2CN_{(s)} + \frac{3}{2}O_2(g) \rightarrow N_{2(g)} + O_{2(g)} + H_2O_{(1)}$ is ____ kJ. (Rounded off to the nearest integer)
[Assume ideal gases and $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

Q3: 25 Feb (Shift 2) - Numerical

Five moles of an ideal gas at 293 K is expanded isothermally from an initial pressure of 2.1MPa to 1.3 MPa against at constant external 4.3 MPa. The heat transferred in this process is ____ kJ mol^{-1}. (Rounded-off of the nearest integer) [Use $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$]

Q4: 26 Feb (Shift 1) - Numerical

For a chemical reaction $A + B \rightleftharpoons C + D$ $\left(\Delta_r H^{\ominus} = 80 \text{ kJ mol}^{-1}\right)$ the entropy change $\Delta_r S^{\ominus}$ depends on the temperature T (in K) as $\Delta_r S^{\ominus} = 2T \left(J \text{ K}^{-1} \text{ mol}^{-1}\right)$

Minimum temperature at which it will become spontaneous is ____ K.

Answer Key

Q1 (855) **Q2** (741) **Q3** (15) **Q4** (200)

Q1 (855)

$3HC \equiv CH(g) \rightleftharpoons C_6H_6(\ell)$

$\Delta G_r^\circ = \Delta G_f^\circ [C_6H_8(\ell)] - 3 \times \Delta G_f^\circ [HC \equiv CH]$

$= \left[-1.24 \times 10^5 - 3 \times \left(-2.04 \times 10^5\right)\right]$

$= 4.88 \times 10^5\ \mathrm{J/mol}$

$\Delta G_r^\circ = -RT \ln(K_{eq})$

$\log(K_{eq}) = \frac{-\Delta G^\circ}{2.303RT}$

$= \frac{-4.88 \times 10^5}{2.303 \times 8.314 \times 298}$

$= -8.55 \times 10^1 = 855 \times 10^{-1}$

Q2 (741)

$NH_2CN(s) + \frac{3}{2}O_2(g) \longrightarrow N_2(g) + CO_2(g) + H_2O(\ell)$

$\Delta ng = (1 + 1) - \frac{3}{2} = \frac{1}{2}$

$\Delta H = \Delta U + \Delta ngRT$

$= -742.24 + \frac{1}{2} \times \frac{8.314 \times 298}{1000}$

$= -742.24 + 1.24$

$= 741\ \mathrm{kJ/mol}$

Q3 (15)

Moles $(n) = 5$

$T = 293\mathrm{k}$

Process $=$ IsoT. $\rightarrow$ Irreversible

$P_{\text{ini}} = 2.1$ M Pa

$P_t = 1.3$ M Pa

$P_{\text{ext}} = 4.3$ mPa

Work $= -P_{\text{ext}}\, \Delta v$

$= -4.3 \times \left(\frac{5 \times 293R}{1.3} - \frac{5 \times 293}{2.1}\right)$

$= -5 \times 293 \times 8.314 \times 43 \left(\frac{1}{13} - \frac{1}{21}\right)$

$= \frac{5\times293\times8.314\times43\times8}{21\times13}$

= -15347.7049 J $= -15.34\text{KJ}$

Isothermal process, so $\Delta U = 0$ $w = -Q$

$Q = 15.34\text{KJ/mol}$

So answer is 15

Q4 (200)

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

To make the process spontaneous

$\Delta G^\circ < 0$

$\Delta H^\circ - T\Delta S^\circ < 0$

$T > \frac{\Delta H^\circ}{\Delta S^\circ}$

$T > \frac{80000}{2\ T}$

$2T^2 > 80000$

$T^2 > 40000$

$T > 200$

The minimum temperature to make it spontaneous is 200 K.